Aastha Mittal

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Education

Carnegie Mellon University — Pittsburgh, PA

Bachelor of Science in Computer Science — *Expected May 2028*

- Coursework: Data Structures and Algorithms, Distributed Systems, Parallel Algorithms, Computer Systems, Probability and Statistics, Machine Learning

Experience

IBM Research / HashiCorp R&D — Durham, NC

Software Engineering Intern, Infrastructure and Security — *May 2026 – Jul. 2026*

- Engineered Go backend workflows translating natural-language requirements into Terraform and Vault policy configurations, integrating LLM-generated outputs with infrastructure and authorization services.
- Built an OPA/Rego verification layer with adversarial test cases to validate generated configurations, detect incorrect permissions, and strengthen error handling across security-sensitive infrastructure workflows.
- Prototyped a multi-agent red-team workflow for LLM-generated infrastructure policies, using adversarial evaluation, output validation, and end-to-end tests to investigate failure modes.

Time2Cash — Remote

Software Engineering Intern — *Jan. 2026 – May 2026*

- Built and maintained Python microservices using Kafka for asynchronous processing, PostgreSQL for persistent records, and Redis for fast access, supporting 50K+ legal documents per day at 99.2% uptime.
- Built an LLM and FAISS retrieval pipeline for unstructured financial and legal records, reducing manual review by 65% and surfacing $2M+ in potential revenue leakage through data-driven analysis.
- Investigated data-quality and cross-service processing failures, strengthening input validation, retrieval checks, and test coverage to make document-analysis results more dependable.

Jane Street – INSIGHT Program — New York, NY

Selected Participant — *2026*

- Selected for Jane Street INSIGHT; explored quantitative problem solving and engineering trade-offs through collaborative technical exercises.

Nationwide Insurance — Columbus, OH

Software Engineering Intern — *May 2025 – Aug. 2025*

- Developed Java services and SQL-backed underwriting workflows supporting an enterprise customer base of 1M+, contributing to improvements of 15% in decision accuracy and 40% in user confidence.
- Traced defects across application and database layers, checked SQL data flows, and collaborated on testing and code review to improve the correctness of underwriting-facing features.

Axway Inc. — Remote

Software Engineering Intern, Frontend Development — *Jun. 2023 – Jul. 2023*

- Developed reusable React components for an enterprise API Management Portal, integrating developer-facing interfaces with API-backed application workflows.

Carnegie Mellon University — Pittsburgh, PA

Teaching Assistant, Data Structures and Algorithms — *Aug. 2025 – Present*

- Guide 25+ students weekly through Java debugging, data-structure implementations, and algorithm analysis; explain error diagnosis, correctness, and runtime trade-offs in individual and group sessions.

Projects

Distributed Key-Value Store | *Go, Raft, Concurrency*

- Developed concurrent Go networking and state-management components using goroutines, channels, and RPC, including Raft state inspection and election-related interfaces.
- Tested concurrent message handling and failure-sensitive paths, analyzing leader election behavior, consistency requirements, and recovery trade-offs.

Limit Order Book | *C++, Data Structures, Testing*

- Implemented a C++ limit order book with price-priority matching for buy and sell orders, emphasizing efficient updates, deterministic execution, and correct state transitions.
- Tested partial fills, price-level updates, and boundary conditions to validate matching invariants and state correctness.

Transformer from Scratch | *Python, PyTorch*

- Implemented attention, tensor transformations, and training components for a transformer in PyTorch; debugged shape mismatches and checked forward-pass behavior to understand model execution end to end.

Technical Skills

Languages: Python, Go, C++, Java, SQL, TypeScript

Engineering: Data structures, algorithms, distributed systems, backend development, API integration, automated testing, LLM evaluation

Tools: Kafka, PostgreSQL, Redis, Terraform, Vault, OPA/Rego, FAISS, PyTorch, Git, React